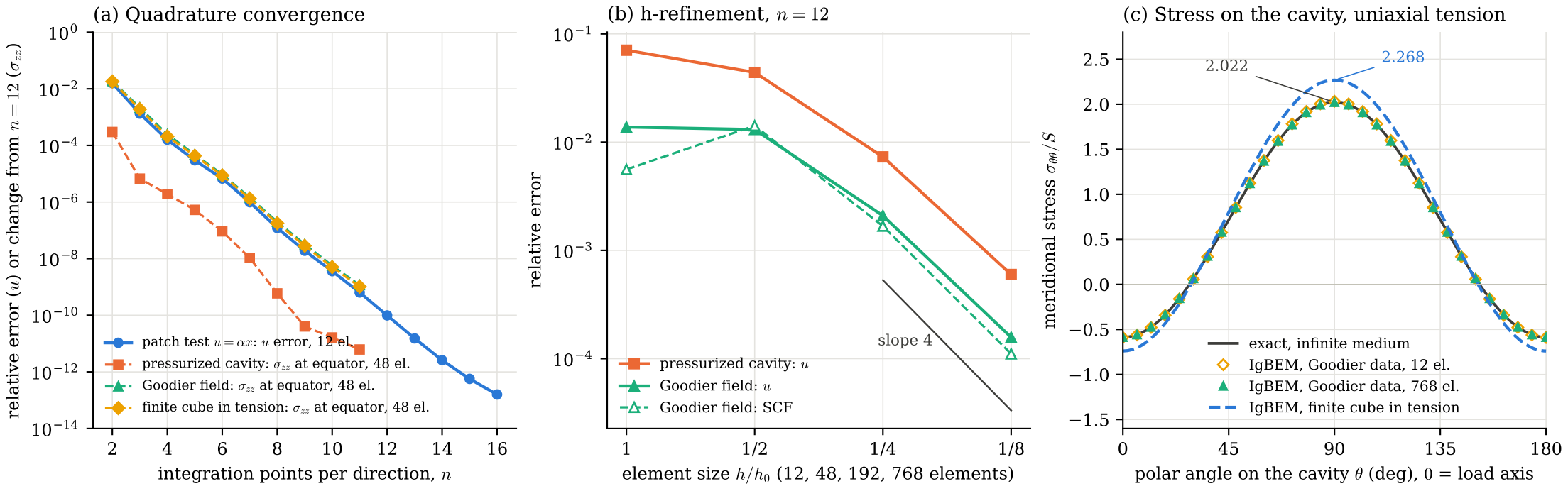
(a) Quadrature convergence
relative error (u) or change from $n=12$ (σ_{zz})
integration points per direction, n
patch test $u=\alpha x$: u error, 12 el.
pressurized cavity: σ_{zz} at equator, 48 el.
Goodier field: σ_{zz} at equator, 48 el.
finite cube in tension: σ_{zz} at equator, 48 el.
(b) h-refinement, $n=12$
relative error
element size h/h_0 (12, 48, 192, 768 elements)
pressurized cavity: u
Goodier field: u
Goodier field: SCF
slope 4
(c) Stress on the cavity, uniaxial tension
meridional stress $\sigma_{\theta\theta}/S$
polar angle on the cavity θ (deg), 0 = load axis
2.022
2.268
exact, infinite medium
IgBEM, Goodier data, 12 el.
IgBEM, Goodier data, 768 el.
IgBEM, finite cube in tension